Lab 3 Naive BST

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I. Lab structure

本实验包含以下文件结构：

```
Lab3 Naive BST/
├── README.md
├── bst_node.h                  # BST node类的定义，Tree类的定义
├── bst_operation.h             # BST有关的操作(题目要求的实现均在此)
├── insert_walk_test.cpp        # 测试insert和前/中/后序遍历实现的正确性
├── search_successor_test.cpp   # 测试后继和前驱查找的正确性
└── search_delete_test.cpp      # 测试查找和删除的正确性
```

主要类说明：

- **node**: 二叉搜索树节点类

```
template <typename T>
class node{
    public:
    T value;
    node* parent;
    node* left_child;
    node* right_child;

    node(T val, node* p=nullptr, node* lc=nullptr, node* rc=nullptr){
        value = val;
        parent = p;
        left_child = lc;
        right_child = rc;
    }
};
```

- **Tree**: 二叉搜索树类

```
template <typename T>
class tree{
    public:
    node<T>* root;
    tree(){
        root = nullptr;
    }
};
```

II. BST methods in `bst_operation.h`

- Insert

```
template <typename T>
void tree_insert(tree<T>& t, T new_val)
```

- **Search(Question1)**

```
template <typename T>
node<T>* tree_search(tree<T> t, T key, bool is_recursion=false)
```

- **后继/前驱(Question2)**

```
template <typename T>
node<T>* successor(node<T>* source)


template <typename T>
node<T>* predeccessor(node<T>* source);
```

- **前/中/后序遍历(Question3)**

```
template <typename T>
vector<T> preorder_walk(tree<T>& t);


template <typename T>
vector<T> inorder_walk(tree<T>& t);


template <typename T>
vector<T> postorder_walk(tree<T>& t);
```

- delete

```
template <typename T>
void tree_delete(tree<T>& t, node<T>* target);
```

III.Test

```
.\search_successor_test
```

```
.\insert_walk_test
```

```
.\search_delete_test
```